

LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 2
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1. TITLE

Design and Implementation of Modern Responsive Webpage Using Advanced CSS3 Properties and Flexbox Layout

2. AIM

To develop a modern, responsive web page incorporating CSS3 linear gradients, text-shadows, flexible box (flexbox) card layout, smooth hover transitions, pill-shaped buttons, and styled footer components.

3. OBJECTIVE

The objectives of this experiment are:

- To apply custom typography (Georgia, Arial, Verdana) and background color palettes.
 - To design a hero header section with horizontal linear gradient (linear-gradient(to right, ...)) and text drop-shadows.
 - To construct a responsive navigation bar with interactive hover background color and box-shadow transitions.
 - To build an equal-width 3-card layout utilizing CSS3 Flexbox (display: flex; flex: 1; gap: 25px).
 - To implement interactive pill-shaped buttons with translateY lift animations on hover.
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4. THEORY

CSS3 provides powerful visual layout capabilities that enhance web user experience. Linear gradients (linear-gradient(to right, #6a11cb, #2575fc)) enable smooth multi-color background transitions without external image assets. Text shadows (text-shadow: 3px 3px 6px ...) add depth and legibility to header headings.

CSS3 Flexbox (display: flex) offers an efficient way to align and distribute space among container elements. Properties like flex: 1 ensure all child card elements automatically maintain equal width and height. Transitions (transition: all 0.3s ease) combined with transform: translateY(-3px) create smooth micro-interactions on button hover.

Application Flow:

HTML5 Document Structure -> CSS3 Gradient Header & Typography -> Navigation Bar Hover Transitions -> Flexbox Card Layout Construction -> Pill Button Hover Transformations -> Styled Footer Rendering

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document structure containing <header>, <nav>, <main class='card-container'> with 3 card items, and <footer>.
2. Apply CSS3 reset rules (margin: 0; padding: 0) and set base body font to Arial with background color #f4f5f7.
3. Style the header with horizontal linear-gradient (#6a11cb to #2575fc), white Georgia serif title (3.5rem), and text-shadow (3px 3px 6px).
4. Style navigation links with dark background (#333), Verdana font, rounded borders, and hover state (#2575fc background with box-shadow).
5. Configure .card-container with display: flex, justify-content: center, gap: 25px, and max-width: 1100px.
6. Style each .card with white background, 2px solid #ddd border, border-radius 12px, padding 25px, box-shadow (0 8px 16px), and flex: 1.
7. Format card headings (h2), description paragraphs (p), and pill-shaped action buttons (border-radius: 25px).
8. Implement hover micro-interactions for buttons using transform: translateY(-3px) and box-shadow elevation.
9. Style dark footer section (#222 background, #aaa text) with centered copyright information.
10. Open web page in browser and verify layout alignment, flexbox responsiveness, and hover effects.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My Styled Webpage</title>
  <style>
    /* 1. Different font families and Background colors applied to the body */
    body {
      font-family: 'Arial', sans-serif;
      background-color: #f4f5f7;
      margin: 0;
      padding: 0;
    }

    /* 2. Gradient background and Text Shadow in the Header */
    header {
      background: linear-gradient(to right, #6a11cb, #2575fc);
      color: white;
      text-align: center;
      padding: 40px 20px;
    }

    header h1 {
      font-family: 'Georgia', serif;
      font-size: 3.5rem;
      margin: 0;
      text-shadow: 3px 3px 6px rgba(0, 0, 0, 0.4);
    }

    /* 3. Navigation Bar Styling */
    nav {
      background-color: #333;
      text-align: center;
      padding: 15px 0;
    }

    nav a {
      color: white;
      text-decoration: none;
      font-family: 'Verdana', sans-serif;
      font-size: 1.1rem;
      margin: 0 15px;
      padding: 10px 20px;
      border-radius: 5px;
      transition: 0.3s;
    }

    /* 4. Hover effect for navigation links */
    nav a:hover {
      background-color: #2575fc;
      color: #ffffff;
      box-shadow: 0 4px 8px rgba(37, 117, 252, 0.4);
    }

    /* 5. Flexbox Layout for the Content Cards */
    .card-container {
      display: flex;
      justify-content: center;
      gap: 25px;
      padding: 40px 20px;
      margin: 0 auto;
      max-width: 1100px;
    }

    /* 6. Cards: Borders, Border Radius, Box Shadow, Padding */
    .card {
      background-color: white;
      border: 2px solid #ddd;
      border-radius: 12px;
      padding: 25px;
      text-align: center;
    }
```

```

        box-shadow: 0 8px 16px rgba(0, 0, 0, 0.1);
        flex: 1;
    }

    .card h2 {
        font-size: 1.8rem;
        color: #333;
        margin-bottom: 15px;
    }

    .card p {
        font-size: 1rem;
        color: #666;
        line-height: 1.5;
        margin-bottom: 25px;
    }

    /* 7. Button Styling */
    .card button {
        background-color: #6a11cb;
        color: white;
        border: none;
        padding: 10px 25px;
        font-size: 1rem;
        border-radius: 25px;
        cursor: pointer;
        transition: all 0.3s ease;
    }

    /* 8. Hover effect for buttons */
    .card button:hover {
        background-color: #2575fc;
        transform: translateY(-3px);
        box-shadow: 0 6px 12px rgba(37, 117, 252, 0.3);
    }

    /* 9. Footer Styling */
    footer {
        background-color: #222;
        color: #aaa;
        text-align: center;
        padding: 20px;
        font-size: 0.9rem;
        margin-top: 20px;
    }
</style>
</head>
<body>

    <header>
        <h1>My Styled Webpage</h1>
        <p>A demonstration of modern CSS properties</p>
    </header>

    <nav>
        <a href="#home">Home</a>
        <a href="#about">About</a>
        <a href="#services">Services</a>
        <a href="#contact">Contact</a>
    </nav>

    <main class="card-container">

        <div class="card">
            <h2>Web Design</h2>
            <p>Creating beautiful, responsive interfaces using HTML5 and modern CSS techniques like flexbox and grid layouts.</p>
            <button>Learn More</button>
        </div>

        <div class="card">
            <h2>Graphics</h2>
            <p>Utilizing the HTML5 Canvas element to draw shapes, text, and dynamic graphics directly in the browser.</p>

```

```
        <button>Learn More</button>
    </div>

    <div class="card">
        <h2>Interactivity</h2>
        <p>Bringing web pages to life with hover effects, transitions, smooth shadows, and interactive UI
elements.</p>
        <button>Learn More</button>
    </div>

</main>

<footer>
    <p>&copy; 2026 My Styled Webpage. All CSS requirements met successfully.</p>
</footer>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
linear-gradient(to right, #6a11cb, #2575fc)	Renders a smooth horizontal gradient background transition in header.
text-shadow: 3px 3px 6px rgba(0,0,0,0.4)	Applies a 3D drop-shadow effect to header title text for visual depth.
display: flex; gap: 25px	Formats card elements into a responsive flexbox row layout with 25px spacing.
flex: 1	Instructs all card components to expand equally to fill container width.
border-radius: 12px	Rounds the corners of card boxes for modern UI appearance.
box-shadow: 0 8px 16px rgba(0,0,0,0.1)	Adds subtle elevation shadows behind content card elements.
border-radius: 25px	Creates pill-shaped action buttons with fully rounded ends.
transform: translateY(-3px)	Elevates buttons by 3px upward on mouse hover for interactive feedback.

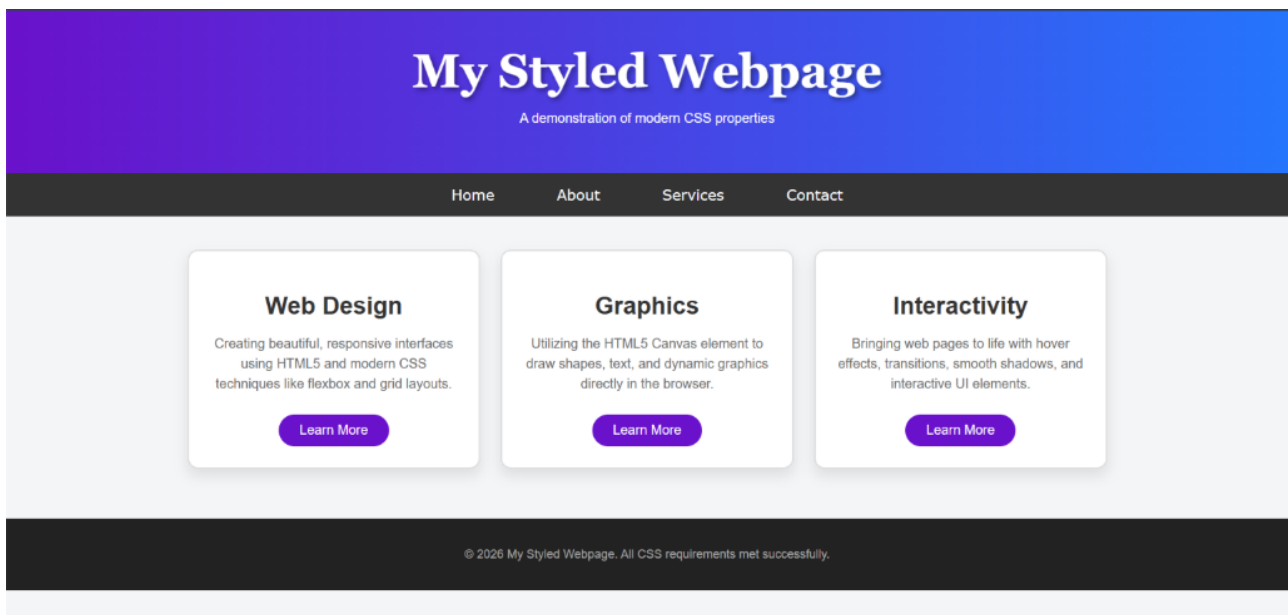
8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Gradient Header Render	Open webpage in browser	Purple-to-blue gradient header with drop shadow title displays	Pass
TC-02	Nav Link Hover Effect	Hover over navigation links	Background changes to #2575fc with glowing box shadow	Pass
TC-03	Flexbox Card Alignment	Inspect content container	3 equal-width cards (Web Design, Graphics, Interactivity) align in row	Pass
TC-04	Button Hover Animation	Hover over 'Learn More' buttons	Button lifts up -3px, color turns blue with shadow elevation	Pass
TC-05	Footer Rendering	Scroll to page bottom	Dark footer with copyright statement renders cleanly across full width	Pass

9. OUTPUT

Output 1: Modern CSS Webpage with Gradient Header, Navigation Bar, Flexbox Cards & Footer



10. RESULT

Thus, the modern webpage demonstrating advanced CSS3 properties including linear gradients, flexbox layout, card components, text shadows, and hover transitions was successfully designed, implemented, and verified.